

Flashcards — Chemistry Chapter 6: Stoichiometry

How to use these cards: Cover the right column with your hand or a piece of paper. Read the question, say your answer out loud, then slide down to check. If you get it wrong, put a dot next to it — come back and re-test the dots at the end.

Daily target: 15 minutes of flashcards each day. Mix the order so your brain has to work harder.

Cards are organized by **topic** and **difficulty**: - **Easy** — facts and definitions (you should get 100%) - **Medium** — apply a rule or do a quick calculation - **Hard** — multi-step thinking or a trickier distinction

Topic 1 — Empirical vs Molecular Formula

Easy

Q	A
What is an empirical formula?	The simplest whole-number ratio of atoms in a compound.
What is a molecular formula?	The actual number of each type of atom in one molecule of a compound.
What does "empirical" mean in the phrase "empirical formula"?	Based on experiment or experience.
Give one example of a compound where molecular and empirical formulas are the same.	H ₂ O (also CO ₂ , NH ₃ , CH ₄).
Molecular formula of glucose?	C ₆ H ₁₂ O ₆

Medium

Q	A
Empirical formula of glucose (C ₆ H ₁₂ O ₆)?	CH ₂ O
Empirical formula of benzene (C ₆ H ₆)?	CH
Empirical formula of hydrogen peroxide (H ₂ O ₂)?	HO
Empirical formula of ethene (C ₂ H ₄)?	CH ₂
Compound has 2 C, 4 H, 2 O. What is its empirical formula?	CH ₂ O

Hard

Q	A
How do you find an empirical formula from a molecular formula?	Divide all subscripts by their highest common factor (HCF).
If a compound has molecular formula C ₆ H ₁₂ O ₆ and empirical formula CH ₂ O, what must its molecular mass be a multiple of?	30 (the mass of the empirical unit).
Two compounds have the same empirical formula (CH ₂ O). What's the simplest way they could still be different compounds?	Different molecular masses — one might be CH ₂ O, another could be C ₂ H ₄ O ₂ (acetic acid) or C ₆ H ₁₂ O ₆ (glucose).

Topic 2 — Molecular Mass & Formula Mass

Easy

Q	A
What is molecular mass?	The sum of atomic masses of all atoms in one molecule.
Unit of molecular mass?	amu (atomic mass unit).
What is formula mass?	The sum of atomic masses of all atoms in one formula unit of an ionic compound.
Which term is used for ionic compounds: molecular mass or formula mass?	Formula mass.
Atomic mass of hydrogen?	1 amu (1.008 amu exactly).
Atomic mass of oxygen?	16 amu
Atomic mass of carbon?	12 amu
Atomic mass of sodium?	23 amu
Atomic mass of chlorine?	35.5 amu

Medium

Q	A
Molecular mass of H ₂ O?	18 amu
Molecular mass of CO ₂ ?	44 amu
Molecular mass of NH ₃ ?	17 amu
Molecular mass of CH ₄ ?	16 amu
Formula mass of NaCl?	58.5 amu
Formula mass of CaCO ₃ ?	100 amu
Formula mass of Mg(OH) ₂ ?	58 amu
Formula mass of H ₂ SO ₄ ?	98 amu
Formula mass of KClO ₃ ?	122.5 amu
Formula mass of glucose (C ₆ H ₁₂ O ₆)?	180 amu

Hard

Q	A
Formula mass of CuSO ₄ ·5H ₂ O (Cu=63.5, S=32, O=16, H=1)?	249.5 amu
A compound Na ₂ CX ₃ has formula mass 106 amu. What is the atomic mass of X?	12 (it's sodium carbonate Na ₂ CO ₃).
Why are molecular and formula mass calculated the same way even though they have different names?	Because both are just "sum of atomic masses of every atom in the formula" — the calculation doesn't change. The name just describes whether the compound is molecular or ionic.

Topic 3 — The Mole & Avogadro's Number

Easy

Q	A
What is a mole?	The amount of a substance that contains 6.022×10^{23} particles.
What is Avogadro's number?	6.022×10^{23}
Symbol for Avogadro's number?	N_A
Who is Avogadro's number named after?	Amedeo Avogadro, an Italian chemist.
1 dozen = ?	12
1 mole = ? particles	6.022×10^{23}

Medium

Q	A
How many carbon atoms in 1 mole of carbon?	6.022×10^{23}
How many water molecules in 1 mole of water?	6.022×10^{23}
How many molecules in 2 moles of CO ₂ ?	$2 \times 6.022 \times 10^{23} = 1.2044 \times 10^{24}$
How many molecules in 0.5 moles of O ₂ ?	$0.5 \times 6.022 \times 10^{23} = 3.011 \times 10^{23}$
How many atoms in 0.25 moles of Na?	$0.25 \times 6.022 \times 10^{23} = 1.506 \times 10^{23}$

Hard

Q	A
A compound has 3.011×10^{23} molecules. How many moles is that?	0.5 moles (it's half of Avogadro's number).
Why did chemists invent the mole instead of just counting atoms?	Atoms are too small to count individually. The mole is a convenient counting unit that makes the numbers practical.
If one mole of a substance contains 6.022×10^{23} particles, how many particles in 0.001 mole?	6.022×10^{20}

Topic 4 — Molar Mass

Easy

Q	A
What is molar mass?	The mass of one mole of a substance, in grams.
What is gram atomic mass?	The atomic mass of an element expressed in grams (= 1 mole of that element).
What is gram molecular mass?	The molecular mass of a compound expressed in grams.
What is gram formula mass?	The formula mass of an ionic compound expressed in grams.
What is the formula for moles from mass?	$n = \text{mass} \div \text{molar mass}$
Mass of 1 mole of carbon?	12 g
Mass of 1 mole of water?	18 g
Mass of 1 mole of NaCl?	58.5 g

Medium

Q	A
How many moles in 36 g of water?	$36 \div 18 = 2$ moles
How many moles in 44 g of CO ₂ ?	$44 \div 44 = 1$ mole
How many moles in 11 g of CO ₂ ?	$11 \div 44 = 0.25$ moles
Mass of 0.5 moles of NaCl?	$0.5 \times 58.5 = 29.25$ g
Mass of 2 moles of water?	$2 \times 18 = 36$ g
Mass of 0.25 moles of CO ₂ ?	$0.25 \times 44 = 11$ g
Mass of 1.5 moles of H ₂ SO ₄ ?	$1.5 \times 98 = 147$ g

Hard

Q	A
How many molecules are present in 0.25 moles of SO ₂ ?	$0.25 \times 6.022 \times 10^{23} = 1.5055 \times 10^{23}$ molecules
How many moles are present in 1 billion (10 ⁹) atoms of magnesium?	$10^9 \div 6.022 \times 10^{23} = 1.66 \times 10^{-15}$ moles
How many atoms in 1.25 moles of Zn?	$1.25 \times 6.022 \times 10^{23} = 7.53 \times 10^{23}$ atoms
Why is the molar mass of water (18 g) numerically the same as its molecular mass (18 amu)?	By design — the mole was defined so that atomic mass in amu equals molar mass in grams. This is the bridge between the atomic and macroscopic worlds.

Topic 5 — Binary Ionic Formulas

Easy

Q	A
What is a cation?	A positively charged ion (usually a metal).
What is an anion?	A negatively charged ion (usually a non-metal).
What is the charge of sodium ion?	+1 (Na ⁺)
What is the charge of chloride ion?	−1 (Cl [−])
What is the charge of calcium ion?	+2 (Ca ²⁺)
What is the charge of oxide ion?	−2 (O ^{2−})
What is the charge of aluminium ion?	+3 (Al ³⁺)

Medium

Q	A
Formula of sodium chloride?	NaCl
Formula of potassium oxide?	K ₂ O
Formula of magnesium chloride?	MgCl ₂
Formula of calcium oxide?	CaO
Formula of aluminium oxide?	Al ₂ O ₃
Formula of aluminium chloride?	AlCl ₃
Formula of magnesium nitride (Mg ²⁺ and N ^{3−})?	Mg ₃ N ₂

Hard

Q	A
Why must an ionic compound be electrically neutral?	Because the total positive charge from cations must equal the total negative charge from anions; otherwise the compound would have a net charge.
What's the crisscross method?	Use the charge of one ion as the subscript of the other. For Al^{3+} and O^{2-} , you get Al_2O_3 .
Formula of ammonium sulfate (NH_4^+ and SO_4^{2-})?	$(\text{NH}_4)_2\text{SO}_4$

Topic 6 — Balancing Chemical Equations

Easy

Q	A
What is a reactant?	A substance that starts a chemical reaction (on the left of the arrow).
What is a product?	A substance formed in a chemical reaction (on the right of the arrow).
What does the arrow in a chemical equation mean?	"Becomes" or "reacts to form".
State symbol for solid?	(s)
State symbol for liquid?	(l)
State symbol for gas?	(g)
State symbol for aqueous (dissolved in water)?	(aq)

Medium

Q	A
In balancing, what can you change?	The coefficients (big numbers in front of formulas).
In balancing, what can you NEVER change?	The subscripts inside a chemical formula.
Why can't you change subscripts?	Changing a subscript changes the identity of the compound (e.g., $\text{H}_2\text{O} \rightarrow \text{H}_2\text{O}_2$ is a completely different substance).
Balance: $\text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{O}$	$2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$
Balance: $\text{Na} + \text{Cl}_2 \rightarrow \text{NaCl}$	$2\text{Na} + \text{Cl}_2 \rightarrow 2\text{NaCl}$
Balance: $\text{Mg} + \text{O}_2 \rightarrow \text{MgO}$	$2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$
Balance: $\text{CH}_4 + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$	$\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$

Hard

Q	A
Balance: $\text{Fe} + \text{O}_2 \rightarrow \text{Fe}_2\text{O}_3$	$4\text{Fe} + 3\text{O}_2 \rightarrow 2\text{Fe}_2\text{O}_3$
Balance: $\text{Al} + \text{HCl} \rightarrow \text{AlCl}_3 + \text{H}_2$	$2\text{Al} + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2$
Balance: $\text{C}_3\text{H}_8 + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$ (propane combustion)	$\text{C}_3\text{H}_8 + 5\text{O}_2 \rightarrow 3\text{CO}_2 + 4\text{H}_2\text{O}$
What law of physics requires us to balance equations?	The Law of Conservation of Mass — matter is neither created nor destroyed in a chemical reaction.
Who discovered that law?	Antoine Lavoisier in the 18th century.

Topic 7 — Ionic Equations

Easy

Q	A
What is an ionic equation?	A chemical equation in which aqueous compounds are written as their separated ions.
What is a net ionic equation?	An ionic equation with the spectator ions removed.
What are spectator ions?	Ions that appear unchanged on both sides of an ionic equation and don't participate in the reaction.
Which compounds do you split into ions in an ionic equation?	Only aqueous (aq) compounds.
Which compounds do you NOT split?	Solids (s), liquids (l), and gases (g).

Medium

Q	A
Net ionic equation for: $\text{HCl(aq)} + \text{NaOH(aq)} \rightarrow \text{NaCl(aq)} + \text{H}_2\text{O(l)}$	$\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$
Net ionic equation for: $\text{AgNO}_3\text{(aq)} + \text{NaCl(aq)} \rightarrow \text{AgCl(s)} + \text{NaNO}_3\text{(aq)}$	$\text{Ag}^+ + \text{Cl}^- \rightarrow \text{AgCl(s)}$
In the neutralization equation above, what are the spectator ions?	Na^+ and Cl^-

Hard

Q	A
Why do different acid-base neutralizations have the same net ionic equation?	Because the underlying reaction is always H^+ combining with OH^- to form water. The other ions are spectators.
Who first proposed the theory of ionic dissociation (that salts break into ions in water)?	Svante Arrhenius in 1884. His PhD committee almost failed him for the idea — he won the Nobel Prize for it in 1903.

Topic 8 — Molecular vs Structural Formula

Easy

Q	A
What does a molecular formula tell you?	The number of each type of atom in a molecule.
What does a structural formula show?	How the atoms are connected to each other.
Molecular formula of n-butane?	C_4H_{10}
Structural formula of n-butane?	$\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_3$

Medium

Q	A
Molecular formula of $\text{CH}_3\text{-CH}_2\text{-OH}$?	$\text{C}_2\text{H}_6\text{O}$
Molecular formula of $\text{CH}_3\text{-CO-CH}_3$?	$\text{C}_3\text{H}_6\text{O}$
Molecular formula of $\text{CH}_3\text{-CH}_2\text{-NH}_2$?	$\text{C}_2\text{H}_7\text{N}$

Hard

Q	A
What are isomers?	Compounds that have the same molecular formula but different structural formulas.
Give an example of two compounds with the same molecular formula $\text{C}_2\text{H}_6\text{O}$.	Ethanol ($\text{CH}_3\text{-CH}_2\text{-OH}$) and dimethyl ether ($\text{CH}_3\text{-O-CH}_3$).
Why does the structural formula matter if the molecular formula is the same?	Because different arrangements give different physical and chemical properties — e.g., ethanol is a liquid you can drink, while dimethyl ether is a dangerous gas.

Mixed Review Challenge (shuffle these — no topic hints!)

Try these last. They're a mix of everything. No looking back at other cards — just answer.

#	Question	Answer
1	6.022×10^{23} is called what?	Avogadro's number
2	Sum of atomic masses of all atoms in a molecule?	Molecular mass
3	Simplest whole-number ratio of atoms?	Empirical formula
4	What does "(aq)" mean?	Dissolved in water (aqueous)
5	Moles in 10 g of NaOH (molar mass 40)?	0.25 moles
6	What do we call ions that don't react in a net ionic equation?	Spectator ions
7	Formula of magnesium oxide?	MgO
8	What law requires us to balance equations?	Law of Conservation of Mass
9	Empirical formula of hydrazine N ₂ H ₄ ?	NH ₂
10	Molar mass of H ₂ O?	18 g/mol
11	Molecules in 0.5 moles of N ₂ ?	3.011×10^{23}
12	Who discovered conservation of mass?	Antoine Lavoisier
13	Molecular formula of water?	H ₂ O
14	Formula mass of Na ₂ CO ₃ ?	106 amu
15	What does a coefficient in a balanced equation represent?	The number of formula units (or moles) of that substance in the reaction.

How many did you get right? (track your progress)

Mark yourself: - **90%+**: You're ready for the exam. - **70-89%**: Review the topics where you missed, then try again tomorrow. - **Below 70%**: Go back to [notes.md](#) and [explanations.md](#) for the weak topics. Don't rush — build real understanding first, then come back to flashcards.

Repeat these cards every 2 days for a week, and you'll remember them for life.